Best Practices for Biomedical Research Data Management

Module 5: Data Management Policy
Publisher Policies

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This video will describe the objectives of publisher policies regarding research data related to manuscripts they will publish. It will address the growing implementation of journal publishers requiring data supporting research results be made available.

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Publisher Policies

Retraction Watch

NEJM paper on sleep apnea retracted when original data can't be found

with 4 comments

The authors of a paper in the *New England Journal of Medicine* are retracting it, after being unable to find data supporting a table that required corrections.

The NEW ENGLAND JOURNAL of MEDICINE

Here's the notice:

> As the authors of the article entitled "CPAP for the Metabo
> Sleep Apnea"1 published in the Journal on December 15, 2
> errors occurred in the Supplementary Appendix, available
> There were multiple errors in the table on pages 18 and 19
> concerning data on the accumulation of abdominal fat as
> tomography and on carotid intima-media thickness as as
> These errors, in turn, changed some values in Table 4 of t
> alter the conclusions of the article, the primary data could
> from secondary tables. Accordingly, we have no way of co
> wish to retract the article.

The paper, by Surendra K. Sharma of the All India Institute of Medic
69 times, according to Thomson Scientific's Web of Knowledge.

Retraction Watch

"We are living in hell:" Authors retract 2nd paper due to missing raw data

with 15 comments

A 2006 paper investigating the effects of docosahexaenoic acid (DHA) and celecoxib on prostate cancer cells has been retracted because it appears to contain panels that were duplicated, and the authors could not provide the raw data to show otherwise.

IJC International Journal of Cancer

This is the second paper the authors have lost because they couldn't furnish the original data to defend their work against allegations of image manipulation. The reason: the Institute for Cancer Prevention in New York, where the authors did the work, shut its doors abruptly in 2004, co-author Bhagavathi A. Narayanan told us. (The institute closed thanks to $5.7 million in grant that was misspent, the *New York Post* reported at the time.)

Recently, some of Narayanan's papers have been questioned on PubPeer; her work has been the subject of an investigation at New York University, where Narayanan is now based.

Narayanan told us that the criticism of their work has deeply affected her and her co-authors:

http://retractionwatch.com/

Publisher Policies

Journal Data Policies

Do the journals that you publish in have data availability, submission, or retention policies? The top 10 journals where UMMS authors publish and their relevant policies are listed below.

Journal	Publisher	Data Policy
Journal of Biological Chemistry	American Society for Biochemistry and Molecular Biology	Structural data must be included as a supplement to the online paper.
PNAS	National Academy of Science	Authors are encouraged to use SI to show all necessary data. Authors are encouraged to deposit as much of their data as possible in publicly accessible databases.
Journal of Immunology	American Association of Immunologists	High resolution structural data [and] original nucleotide sequences, determined nucleotide sequences encoding reported amino acid sequences, and files of nucleotide sequences derived from high throughput/deep sequencing described in the manuscript must be submitted to the appropriate public database at the time of manuscript submission.
Journal of Virology	American Society for Microbiology	Newly determined nucleotide and/or amino acid sequence data must be deposited and GenBank/ENA/DDBJ accession numbers must be included in the manuscript no later than the modification stage of the review process.
Journal of Cellular Biochemistry	Wiley-Blackwell	n/a
Molecular and Cellular Biology	American Society for Microbiology	Nucleotide/amino acid sequence data, coordinates for new structures of macromoleculesdetermined by x-ray crystallography or cryo-electron microscopy, and all supporting microarray data...
Journal of Cell Biology	Rockerfeller University Press	Author must make all data freely available upon request...
Cell	Elsevier	Mandatory deposition of DNA and protein sequences, biological macromolecules, and microarray data to public repositories; recommended availability of additional data sets...
Biochemistry	American Chemical Society	Protein sequences and structural data must be deposited in a public repository; authors are encouraged to submit circular dichronism and other datasets....
Science	American Association for the Advancement of Science	All data necessary to understand the paper must be made available to any reader in an approved public database or as supplemental materials....

Common Approaches

Publisher	Policy
PLoS	Make all data available without restriction
NPG	Submit data for peer review and make data available to readers
ASM, Elsevier (Cell), ACS	Data deposited to public database with accession numbers included in manuscript
NAS	Encouraged supplemental information and deposit in public repositories

Publisher Policies

Joint Data Archiving Policy

- << Journal >> requires, as a condition for publication, that data supporting the results in the paper should be archived in an appropriate public archive, such as << list of approved archives here <<. Data are important products of the scientific enterprise, and they should be preserved and usable for decades in the future. Authors may elect to have the data publicly available at time of publication, or, if the technology of the archive allows, may opt to embargo access to the data for a period up to a year after publication. Exceptions may be granted at the discretion of the editor, especially for sensitive information such as human subject data or the location of endangered species.

DRYAD

Dryad. Joint Data Archiving Policy (JDAP). Available from: http://datadryad.org/pages/jdap

Publisher Policies

Objectives

- Reproducibility of research
- Validation, replication, reanalysis, reinterpretation, inclusion in meta-analysis
- Reduction of burden on authors
- Easier citation of data, enhancing visibiiltiy and ensuring recognition

PLOS FAQs for Data Policy. Available from:
http://journals.plos.org/plosone/s/data-availability#loc-faqs-for-data-policy